

① Performance in networked Systems & Queuing models

Intro: Loss & delay — What are causes? (Queing)

Delay: Conn. Setup (TCP)

Queing

Serilization

Medium access

Retransmissions

Loss: buffer overflow

Collisions

Checksum Mismatch

(TCP) session drop

Queing Models: $X/Y/C[B]$

X — Arrival Process (M — Markovian)

Y — Service Process (M)

C — # Servers

B — Buffer Size

M/M/1



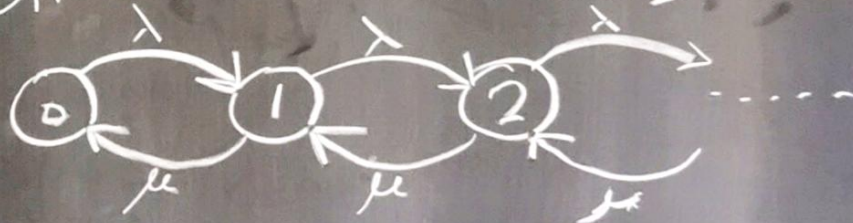
λ — Poisson arrival (rate λ)

μ — exp. distributed Service times (Mean rate μ)

no B = inf buffer

$$\rho = \frac{\lambda}{\mu} \quad \left(\begin{array}{l} \text{Utilization} \\ \rho \geq 1 \text{ unstable} \end{array} \right)$$

Birth-Death Process



λ & μ — Const. don't depend on State

States — Packets in Q (including served)